

Proposal Content

1. Special session title and acronym: **Intelligent Automation (IA)**
2. Brief description (max 500 words):

Intelligent Automation (IA) is revolutionizing modern industries by integrating Artificial Intelligence (AI), robotics, data analytics, and advanced communication technologies to create autonomous, adaptive, and efficient systems. The convergence of Industry 4.0 and the emerging Industry 5.0 paradigm has accelerated the development of human-centric, resilient, and sustainable automation solutions across manufacturing, logistics, healthcare, transportation, energy, and smart infrastructure.

Therefore, this special session aims to bring together researchers, academicians, industry professionals, and practitioners to present recent advances, innovative methodologies, and real-world applications in Intelligent Automation. The session will provide a platform to discuss emerging technologies, address technical challenges, and explore future research directions that enable intelligent, secure, and connected automation systems. The session welcomes the following subtopics:

1. Industry 4.0,
2. Industry 5.0,
3. Digital Twin,
4. Cyber-physical Systems,
5. Cyber security,
6. Smart Manufacturing,
7. IIoT,
8. Warehouse automation
9. Blockchain technology for automation
10. AI applications for automation

3. Names, affiliations, emails, & short CV of the Proposer

- Dr. Koena Mukherjee

Dr. Koena Mukherjee is an Associate Professor in the Department of Electrical and Electronics Engineering at the National Institute of Technology (NIT) Puducherry, India. She obtained her Ph.D. from the Indian Institute of Technology (IIT) Delhi in 2016 and has previously served as a Research Associate at the Indira Gandhi Centre for Atomic Research (IGCAR) and as a faculty member at NIT Silchar.

She has led and collaborated on several government-funded research projects in underwater robotics, marine autonomous systems, and intelligent engineering solutions. Dr. Mukherjee

has authored numerous publications in leading journals, including Ocean Engineering, Robotica, Frontiers in Robotics and AI, and other international journals, and holds multiple granted patents in robotics and intelligent systems. She has received several prestigious recognitions, including the IEEE Robotics and Automation Society (RAS) Developing Country Faculty Engagement Program and the IEEE RAS SPARX Programme, in collaboration with the University of Illinois Urbana-Champaign (UIUC), USA.

She actively contributes to the robotics research community through conference organization, IEEE professional activities, student mentoring, and international collaborations. Her current research focuses on intelligent autonomous systems, marine robotics, biomimetic robots, and next-generation AI-driven automation technologies.

- Dr. Rajvir Kaur

Dr. Rajvir Kaur is an Assistant Professor in the Department of Electrical and Electronics Engineering at the National Institute of Technology (NIT) Puducherry, India. She received her Ph.D. in Electrical Engineering from Malaviya National Institute of Technology (MNIT), Jaipur, in 2019. Following her doctoral studies, she worked as a Postdoctoral Research Fellow at the Singapore University of Technology and Design (SUTD), Singapore, where she conducted research on AI-enabled Cyber Resilience for Power Systems. .

Her research interests encompass electric vehicle powertrains, electric motor drive control techniques, renewable energy integration, energy management systems, and cybersecurity for cyber-physical energy systems. Her work focuses on developing high-performance, reliable, and intelligent power conversion and control solutions for sustainable transportation and smart energy applications.

Dr. Kaur has published numerous research articles in leading international journals and conferences and serves as a reviewer for several reputed IEEE and Elsevier journals. She actively supervises doctoral and postgraduate research and has been involved in organizing faculty development programmes, workshops, and international conferences. She is a Life Member of the Indian Society for Technical Education (ISTE) and a Member of IEEE.

- Dr. Bijukumar B.

Dr. Bijukumar B earned his Doctorate in Electrical Engineering from the National Institute of Technology, Tiruchirappalli, in 2019. Previously, he completed a Master of Engineering in Power Electronics and Drives at Anna University, Chennai, in 2012, and a Bachelor of Technology at Government Engineering College, Idukki, in 2008. Currently, he is an Assistant Professor at the National Institute of Technology Puducherry. His professional

background also includes tenure as an Assistant Engineer (Electrical) for the Kerala State Electricity Board Limited.

His core research focus encompasses power electronic interfaces, renewable energy grid integration, thermoelectric conversion, and model predictive control. Additionally, his work explores energy conversion systems, multilevel converters, advanced control strategies, and the application of Wide Bandgap (WBG) semiconductor devices.

Dr. Bijukumar contributes to the academic community as a reviewer for prestigious publications, including *IEEE Transactions on Industrial Electronics*, *IEEE Journal of Emerging and Selected Topics in Power Electronics*, *IEEE Transactions on Energy Conversion*, and *IEEE Transactions on Power Electronics*. He also evaluates submissions for Elsevier journals such as *Applied Thermal Engineering* and *Energy*.

1. List of potential contributors and paper titles

4.1 Contributors: PG & Ph.D students from Academic Institutions, Faculty members from various organisations, Industry personnel, Scientists from National & International R&D Labs

4.2 Potential paper titles:

- Artificial intelligence driven industrial automation , trends, challenges and future directions.
- Data-Driven Industrial Automation for High-Performance Motion Control Systems
- Intelligent Condition Monitoring and Fault Diagnosis for Industrial Automation Applications
- Industrial Automation Using Intelligent Power Electronic Converters
- Secure Industrial Automation Systems: Integrating Cybersecurity with Industry 4.0
- Industrial Automation Using Collaborative Robots: Opportunities and Challenges

2. Rationale for organizing this session:

The rapid advancement of artificial intelligence (AI), machine learning (ML), robotics, cyber-physical systems, and the Industrial Internet of Things (IIoT) has significantly transformed the landscape of automation across manufacturing, healthcare, transportation, agriculture, logistics, and smart infrastructure. Intelligent automation integrates these technologies to create autonomous, adaptive, and data-driven systems capable of perceiving, reasoning, learning, and making decisions with minimal human intervention. As industries transition toward Industry 5.0, there is a growing need for intelligent systems that not only enhance productivity and efficiency but also improve safety, sustainability, resilience, and human-machine collaboration.

The proposed special session aims to provide an interdisciplinary platform for researchers, academicians, industry practitioners, and policymakers to present recent advances, exchange innovative ideas, and discuss emerging challenges in intelligent automation. The session will foster collaboration between academia and industry by highlighting cutting-edge research in AI-enabled control systems, autonomous robotics, digital twins, intelligent sensing, predictive maintenance, reinforcement learning, edge intelligence, human-robot interaction, and intelligent manufacturing.

By bringing together experts from diverse disciplines, this special session will contribute to advancing the state of the art in intelligent automation while supporting innovation that addresses real-world industrial and societal challenges. The outcomes of the session are expected to stimulate high-quality research collaborations, facilitate knowledge dissemination, and accelerate the adoption of intelligent automation technologies across various sectors.